

IP Addressing and Subnetting -- Guided Notes

Unit 2.3.1 | CompTIA Network+ N10-009 Obj. 2.3 | N10-008 2.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

An IPv4 address is a _____ -bit number divided into four _____ -bit sections called octets. Each octet is written in _____ notation, with values ranging from 0 to _____.

Question 2

Convert the decimal number 172 to binary using the bit-position table (128, 64, 32, 16, 8, 4, 2, 1). Show your work step by step.

Question 3

A subnet mask of 255.255.255.0 written in CIDR notation is _____. The mask tells a device which portion of an IP address is the _____ address and which portion identifies the _____.

Question 4

For the network 192.168.10.0/24, identify: the network address _____, the broadcast address _____, the number of usable host addresses _____, and the valid host range _____.

Question 5

A /26 subnet mask borrows _____ bits from the host portion of a /24. Each /26 subnet provides _____ usable host addresses. How many /26 subnets can be created from a single /24 network?

Question 6

VLSM stands for _____. Unlike fixed-length subnetting, VLSM allows a network engineer to assign _____ -sized subnets to different segments, which prevents _____ of IP address space.

Question 7

Explain the difference between a network address and a broadcast address. Why can neither be assigned to a host? What happens if a packet is sent to the broadcast address?

Question 8 -- Real World Application

REAL WORLD: A network admin needs to allocate subnets for four departments: Engineering (50 hosts), Sales (25 hosts), HR (10 hosts), and a WAN link (2 hosts). Using VLSM starting from 10.0.0.0/24, identify the correct subnet size (/prefix) for each department and explain why you chose it.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-3-1-ip-addressing-and-subnetting/2-3-1-ip-addressing-and-subnetting-answer-key.pdf
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